



Guitar Pedal for Real-Time Lightshows

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Abstract

A digital guitar pedal that, while on and a note is played, will cause a stage light (or lights, if multiple lights are daisy chained together) to quickly flash and fade — serving as a visual parallel to the guitar note as it fades out over time. Whereas most guitar effect pedals allow the user to sculpt the sound of the guitar, the controls of this pedal affect the characteristics of the lights it is connected to, allowing the user to make quick adjustments to the color, brightness, and fade behavior in order to match the mood of their music.



Background

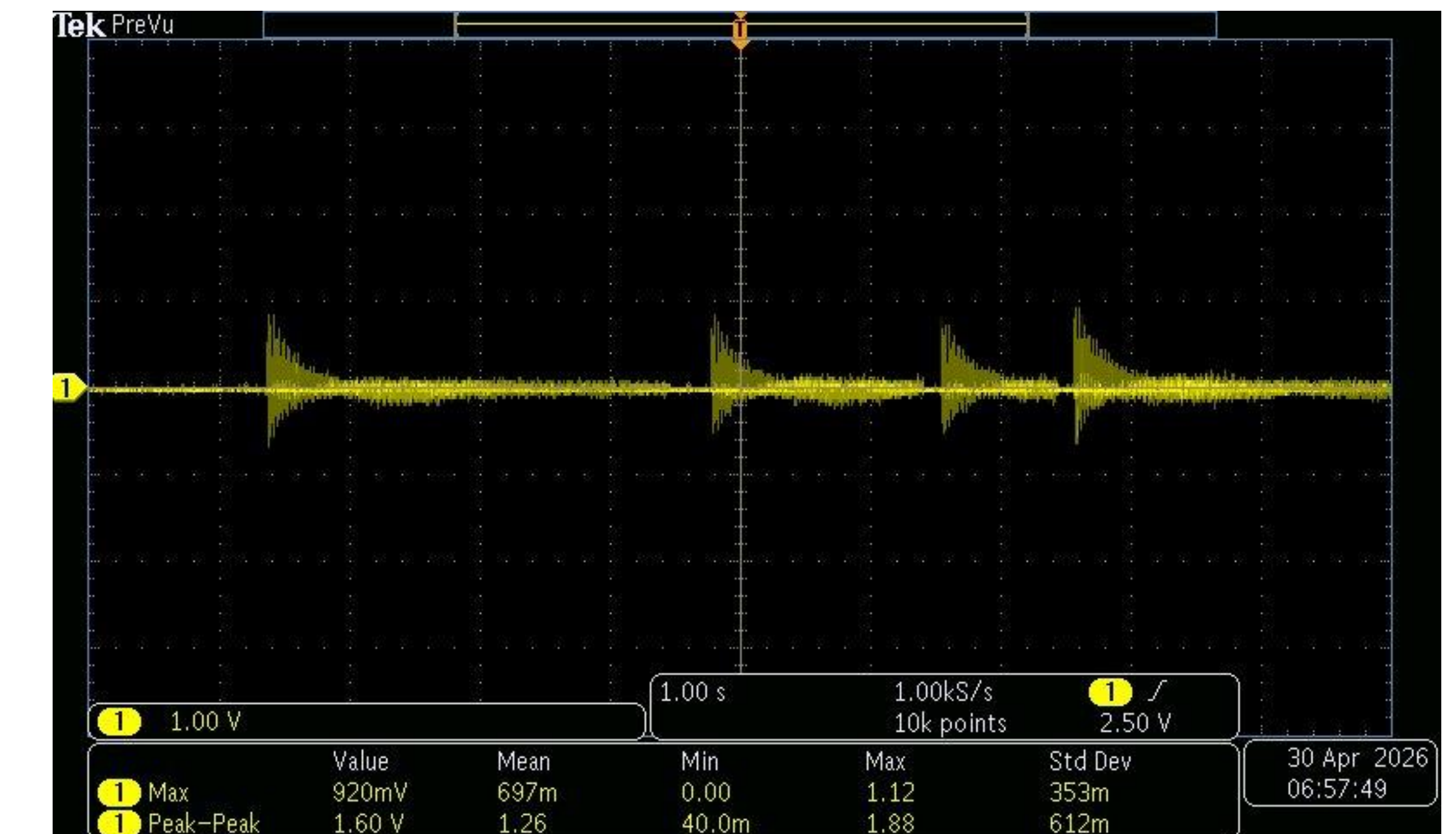
Stage lights are controlled by **DMX**, a unidirectional, packet-based communication protocol in which each DMX “frame” contains up to 512 byte-long “slots”. Each channel of a stage light (R, G, B, W, etc.) is mapped to a slot of the DMX frame. For example, slot 1 controls the red channel values, slot 2 controls the green channel values, etc. DMX frames are transmitted multiple times per second, the speed of which determines the light’s “refresh rate”.

Methods and Materials

Hardware Tools: ST-Link V2, Oscilloscope

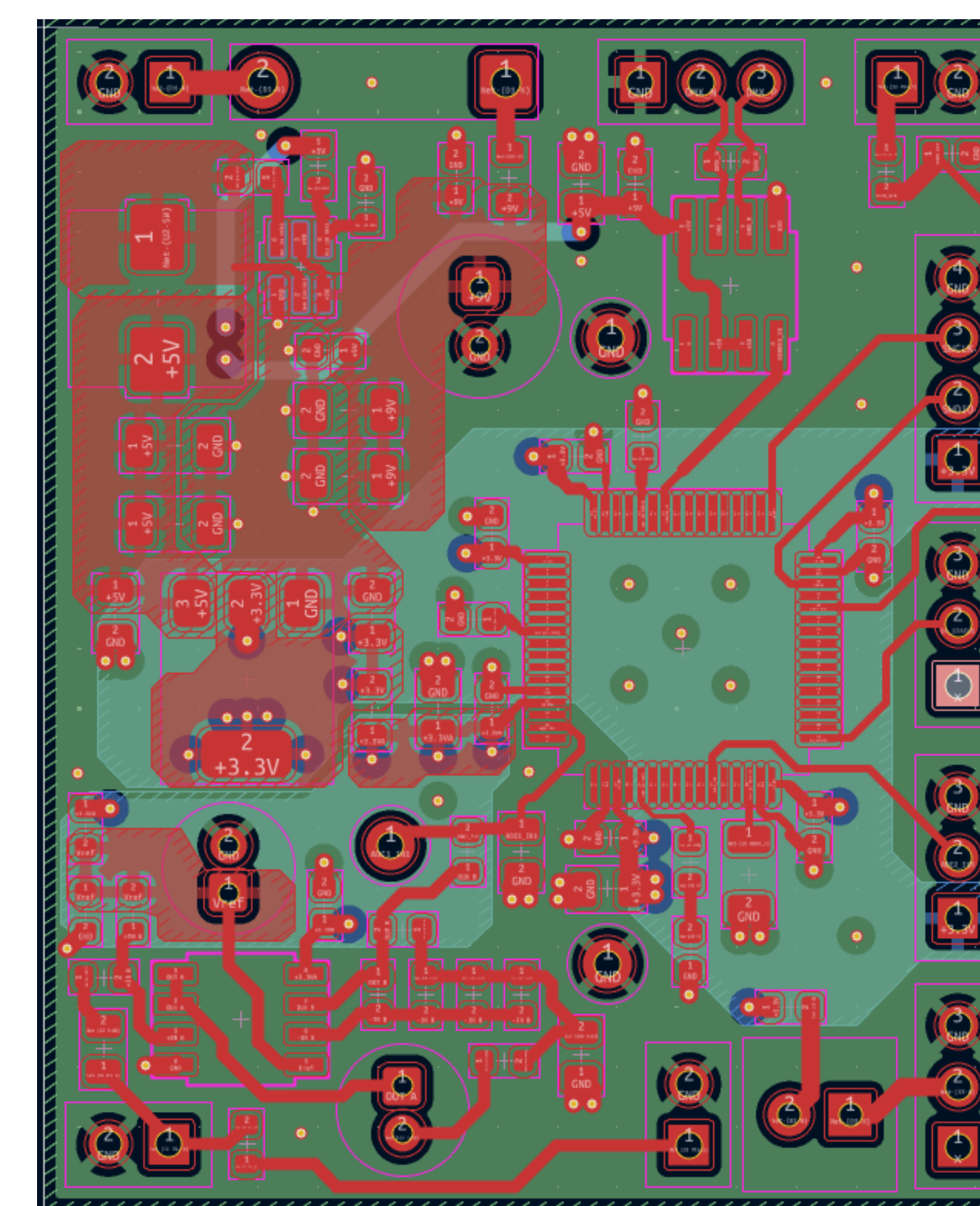
Software Tools: KiCad, STM32CubeIDE, STM32CubeMX, LTspice, GitLab

This pedal is designed around the **STM32F446RE** microcontroller. After the guitar signal is prepared for analog-to-digital conversion (DC biased, buffered, and filtered for aliasing), a peak-picking algorithm is performed on the signal’s envelope to detect the guitar note transients. Each transient will set the selected color channel to the maximum brightness (which corresponds to a value of 255 for that channel’s “slot” in DMX frame) before fading to a value set by the user.



Results

- We were able to succeed in getting the stage light to flash when a note was played on the guitar with **no visible latency**
- The multiple feedback low pass filter has 21 dB of attenuation at the Nyquist frequency (24 kHz)
- As long as the pedal is the first thing in the signal chain, it is compatible with other pedals and amplifiers.



The biggest challenge with our project concerns the stage light itself — we discovered that any jump in brightness while the light is on (such as when the light fading and another note is detected) will cause the stage light's internal firmware to "soften" the brightness change, resulting in a slower, less reactive flash effect.

Future Direction

- Investigate fade behavior with different stage lights
- Try different fade patterns and refresh rates
- refine peak detection algorithm
- Replace unity gain buffer with non-inverting op amp and allow user to fine tune the input gain of the ADC

Acknowledgements

Special thanks to Robert Sweeney in the Engineering and Design department for doing the panel cutouts of our enclosure

